



Geothermal  
Technical Section



Society of Petroleum Engineers

# Geothermal Technical Section

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## NEWSLETTER

**Issue 04**  
April 2026



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## FROM THE SECTION CHAIR

Hello, fellow GTTS members

I am honored to serve as the incoming chair of SPE Geothermal Technical Section.

First, I acknowledge the outstanding achievements of the 2024-2026 Board. Under Reza Khastoo's leadership, several regional membership chairs were created to recruit and serve the members, GTTS newsletters were inaugurated, strong web-based seminars were presented, scholarships awarded, and our Networking Event at ATCE was the best among all Technical Sections. Kudos!!

As GTTS is growing rapidly, I am streamlining the Board organization by creating (a) Director of Technology to coordinate all GTTS Technical Directors and to outreach to other SPE Technical Sections, (b) Global Membership chair to coordinate the efforts of Regional Membership Chairs, and (c) Director of External Engagement to collaborate with other geothermal organizations, such as Geothermal Rising (GR), International Geothermal Association (IGA), Project Innerspace, and GEODE.

This year, SPE Technical Sections will be judged by the following six categories. I will remind myself that these will be my goals.

- (1) Technical Knowledge Sharing,
- (2) Collaboration,
- (3) Section Operations and Sustainability,
- (4) Member Benefits,
- (5) Innovation, and
- (6) Student Integration.

Let us work together to excel both in GTTS and in Geothermal expansion.

**Steve Cheung Ph.D.**  
**Chairperson, SPE Geothermal Technical Section**



# SPE GEOTHERMAL TECHNICAL SECTION

The **SPE Geothermal Technical Section** (SPE GTTS) is a forum for SPE members who share a common interest in geothermal energy and its potential to transform electric power generation and direct use of applications around the world.

The SPE GTTS also focus on the transfer of technology and practices from the oil and gas industry to geothermal energy development and promote industry best practices.

Geothermal Technical Section



## SPE GTTS seeks to:

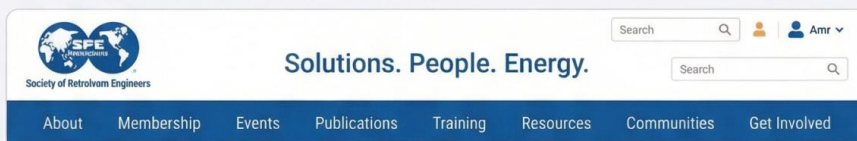
- Collect, disseminate, and exchange technical knowledge concerning geothermal energy, and demonstrate its proximity to more traditional oil and gas disciplines.
- Introduce opportunities to grow careers and businesses by knowledge and skills transfer.
- Establish collaboration across industry and promote industry best practices
- Society of Petroleum Engineers (SPE) and SPE GTTS, are actively engaging through knowledge and technology sharing and transfer to support alternatives and technologies, to a sustainable and low carbon energy transition in the oil and gas industry.



<https://connect.spe.org/gtts/home>

## How to Register for the Geothermal (GTTS) Technical Section

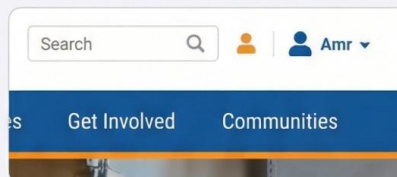
### 1. Go to spe.org



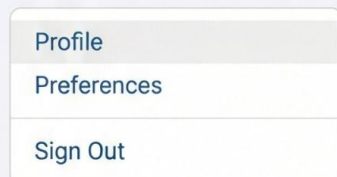
**Supporting Detail:** Visit the official Society of Petroleum Engineers homepage to begin the registration process.

### 2. Login

**Supporting Detail:**  
Access your personal member account by clicking the login icon or your name in the top right corner of the page.



### 3. Click on the 'Profile' option



**Supporting Detail:**  
Open the account dropdown menu and select "Profile" to manage your membership details and preferences.

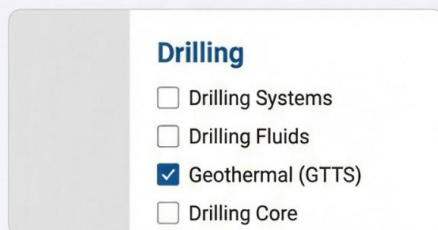
### 4. Scroll down to 'Technical Sections'

**Supporting Detail:**  
Locate the "Preferences" menu on the left-hand sidebar and select the "Technical Sections" link.

#### Preferences

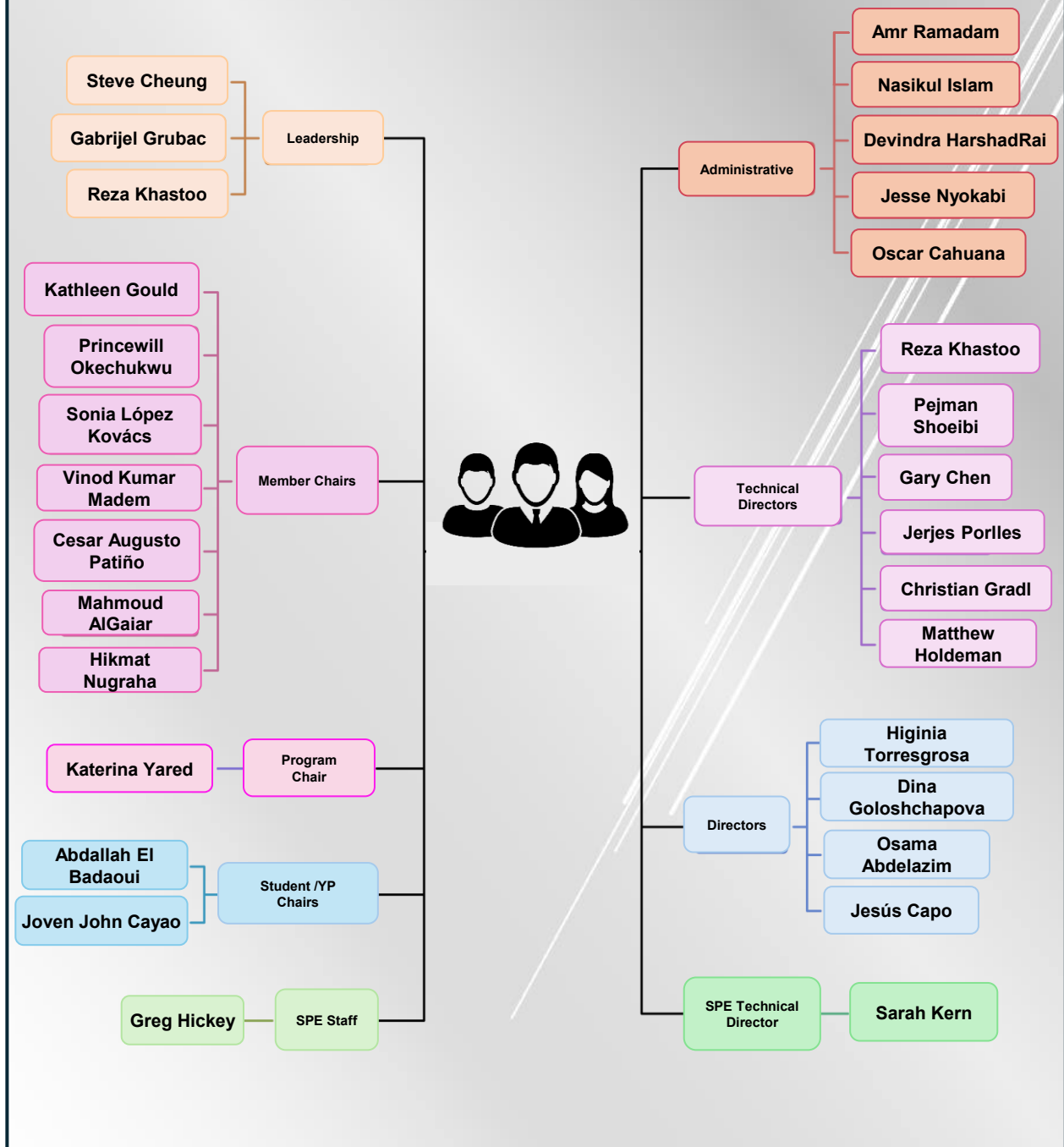
Communication Preferences  
Technical Disciplines  
**Technical Sections**  
Volunteering Interest  
Awards & Recognition  
Mentoring

### 5. Find and select 'Geothermal (GTTS)'



**Supporting Detail:**  
Within the list of technical categories, find the "Drilling" section and check the box for "Geothermal (GTTS)".

## SPE GEOTHERMAL TECHNICAL SECTION OUR TEAM



## Digitalization and AI in the Geothermal Energy Sector: Introduction to a new column in GTTS Newsletter



**by Pejman Shoeibi Omrani**

*Lead scientist at Geological survey of the Netherlands, TNO &  
Data science technical director at SPE GTTS*

### Introducing the column

Geothermal field development is inherently multidisciplinary. Drilling, reservoir engineering, production operations and artificial lift must be coordinated under tight time-frames using geological, geophysical and operational data. As the new Technical Director for GTTS at SPE, I will write a short column each newsletter highlighting digitalization and AI developments in geothermal energy. The column will offer technology digests, interviews with innovators, and summaries of new AI tools and research. My goal is to build a knowledge community around geothermal digitalization and encourage collaboration between AI specialists and geothermal practitioners

### Why digitalization and AI?

Geothermal projects involve complex workflows, from resource assessment and drilling to reservoir management and operations. Decision-making often requires integrating geological, geophysical, reservoir and operational data quickly. AI excels at processing large volumes of structured and unstructured data and can highlight patterns and correlations faster than human experts. The geothermal sector is still emerging. Operators deal with high uncertainties in subsurface properties, complex fluid chemistry and equipment failures. AI-driven analytics and digital twins can help optimize designs, reduce costs and improve reliability. Many geothermal problems require combining physics-based models with data-driven methods. New hybrid AI approaches (e.g., physics-based machine learning) preserve scientific rigor while speeding up simulations and decision-making.

| Discipline               | AI application                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Key benefit/ example                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Drilling                 | AI-driven drilling advisory systems: The OptiDrill project uses machine-learning models and novel sensors to build a real-time drilling advisory system. Modules predict rate of penetration and lithology, detect drilling hazards and optimize well completion.                                                                                                                                                                                                 | Improved drilling efficiency, reduced risks and lower costs.                       |
| Reservoir management     | Physics-based machine learning (PBML): Reservoir simulations are computationally demanding. Researchers at RWTH Aachen and GFZ demonstrated a non-intrusive reduced basis method that couples physics-based models and machine learning to create surrogate models for geothermal reservoirs. These surrogate models capture complex coupled processes while significantly reducing computation time, enabling probabilistic analyses and parameter sensitivities | Faster reservoir simulations and uncertainty quantification for field development. |
| Production and operation | AI-enhanced artificial lift operations: SLB's Intelligent Lift service collects real-time data from artificial-lift systems and uses advanced software and AI to maximize performance and uptime. The digital platform integrates data-driven analytics with physics-based models to simulate pump behavior and optimize operating conditions.                                                                                                                    | Increased production, reduced unplanned shutdowns and optimized power consumption. |
| Data science             | GAIA (Geothermal Analytics and Intelligent Agent): an agentic AI system for geothermal field development. The GAIA framework is a joint development by KAUST and Lawrence Berkley National Lab that integrates three components, GAIA Agent, GAIA Chat and GAIA Digital Twin, in an agentic retrieval-augmented generation workflow.                                                                                                                              | Use of novel AI technologies to automate workflows and processes                   |

## Digitalization and AI in the Geothermal Energy Sector: Introduction to a new column in GTTS Newsletter (Cont.)



**by Pejman Shoeibi Omrani**

*Lead scientist at Geological survey of the Netherlands, TNO &  
Data science technical director at SPE GTTS*

### Trending news

#### **Zanskar raises \$115 million for AI-driven geothermal discovery**

In January 2026, U.S. geothermal developer Zanskar announced a \$115 million Series C funding round to expand its AI-based geothermal discovery platform. Spring Lane Capital led the round, which included Obvious Ventures, Union Square Ventures and Lowercarbon Capital. Zanskar plans to use the capital to scale its AI-native platform, accelerate exploration drilling and begin building geothermal power plants at sites already identified. The company highlights that AI enables unified data management and rapid discovery of geothermal resources, reducing exploration costs and timelines. The new funding allows Zanskar to advance its multi-gigawatt development pipeline and deliver electricity before 2030.

#### **TNO's GEMINI digital twin moves toward open-source release**

TNO's GEMINI project, launched in 2023, is developing an open-source digital twin platform that integrates physics-based, chemistry and data-driven models with real-time data from geothermal plants. The platform aims to provide geothermal operators with a secure, open repository of models and data to monitor, maintain, forecast and optimize system performance. GEMINI digital twin uses advanced simulation tools that combine physics-based and AI models, and it will offer modules for equipment monitoring, model calibration, emission tracking and maintenance scheduling. The framework is being tested in Dutch national programs and will be released as open-source libraries after demonstration. The project is a joint initiative by TNO, WEP, Helin, HVC, Gaia Energy and supported by the Netherlands enterprise agency.

### Final remarks

In future newsletters, this column will continue to explore cutting-edge digital and AI technologies and highlight new research and projects. If you are working on AI tools, digital twin frameworks, or other innovations related to geothermal energy, please contact us to share your insights or discuss collaboration opportunities. Together we can build a more efficient, resilient and sustainable geothermal sector.

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- [Innovation News Network](#)
- [TNO.nl](#)
- [Think GeoEnergy](#)
- [SLB](#)

## Implementation of the World's First >300 °C Propped Enhanced Geothermal System: Lessons Learned and Implications for Commercial Superhot Rock Development



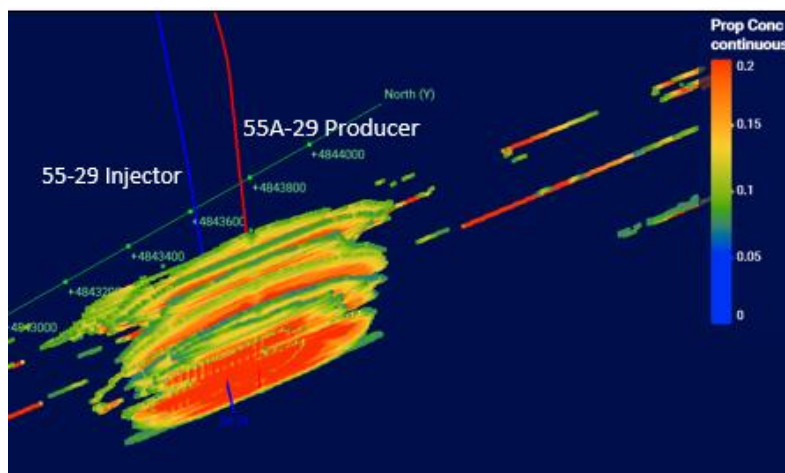
by **Gabrijel Grubac**

*Senior Completion Engineer Mazama Energy, & Vice-Chair at SPE GTTS*

Enhanced Geothermal Systems (EGS) represent one of the most promising pathways to unlock scalable, baseload geothermal energy globally, particularly from superhot rock resources exceeding 300 °C. However, these environments present significant technical challenges, including extremely low natural permeability, harsh thermal conditions, and the need to engineer durable fracture networks capable of sustaining long-term circulation. The Newberry Volcano project in Oregon represents a major milestone in geothermal engineering, successfully implementing the world's first propped EGS reservoir at temperatures exceeding 300 °C. This achievement demonstrates the feasibility of adapting advanced stimulation, completion, and diagnostic technologies to enable reliable reservoir creation and connectivity in superhot rock environments.

The Newberry EGS pilot involved stimulation of an existing injector well and the drilling and stimulating of a parallel producer well to create a connected reservoir system at approximately 3 km depth, where reservoir temperatures exceeded 330 °C. Due to the extremely low natural permeability of the volcanic formations, hydraulic stimulation was required to create conductive pathways capable of supporting fluid circulation and heat extraction. A hybrid stimulation approach was implemented, combining high-viscosity friction reducers with crosslinked CMHPG-based fluids to ensure effective proppant transport and fracture conductivity under extreme thermal conditions. The wells were placed approximately 90–100 meters apart, based on fracture modeling, stress analysis, and connectivity probability simulations, maximizing the likelihood of creating a connected engineered reservoir.

Advanced diagnostics played a critical role in both the execution and validation of the stimulation program. Permanent fiber-optic distributed acoustic sensing (DAS) and distributed temperature sensing (DTS) systems were deployed to monitor fracture propagation and reservoir response in real time. These systems provided unprecedented visibility into subsurface processes, enabling confirmation of fracture growth direction, thermal response, and hydraulic communication between wells. In addition, nanoparticle tracers and chemical tracers were deployed to independently verify connectivity and characterize flow pathways within the engineered reservoir. Microseismic monitoring provided additional confirmation of fracture creation and growth, demonstrating alignment with the regional stress regime and validating fracture modeling predictions.



*Modelled iterations of fracture extensions from producer to injector well*

## Implementation of the World's First >300 °C Propped Enhanced Geothermal System: Lessons Learned and Implications for Commercial Superhot Rock Development



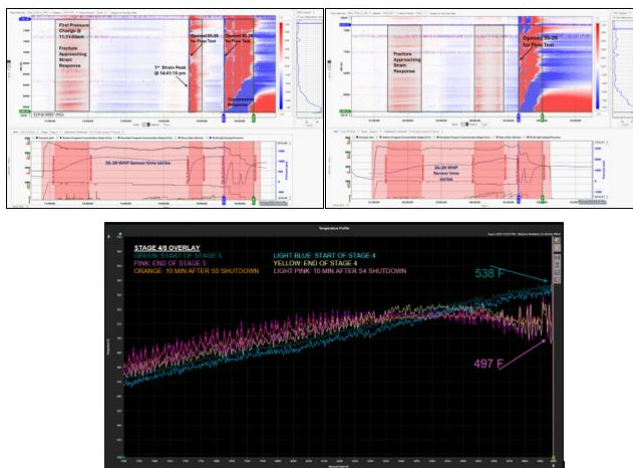
**by Gabrijel Grubac**

*Senior Completion Engineer Mazama Energy, & Vice-Chair at SPE GTTS*

The stimulation program achieved several historic technical milestones. Propped fractures were successfully created and extended between the injector and producer wells, with fracture half-lengths exceeding 80–110 meters. Physical confirmation of proppant transport between wells was obtained through core analysis, providing direct evidence of fracture connectivity and engineered reservoir creation. Tracer testing further confirmed hydraulic communication between wells, with tracer breakthrough observed within hours of injection and measured residence times demonstrating effective fluid circulation within the fracture network. Fiber-optic diagnostics recorded real-time strain and temperature responses in the offset well during stimulation, providing clear confirmation of fracture connection and fluid transmission between wells.

The project also demonstrated the successful deployment of hybrid stimulation technologies specifically adapted for extreme geothermal environments. The hybrid fluid system provided sufficient viscosity and thermal stability to transport proppant effectively under superhot rock conditions, while hybrid completion methods incorporating sliding sleeves and plug-and-perforation systems enabled effective fracture initiation and placement. These results represent a major advancement in geothermal stimulation engineering and demonstrate that proppant-based fracture stimulation is feasible and effective at temperatures exceeding 300 °C.

Equally important were the diagnostic and monitoring advances demonstrated during the project. Fiber-optic monitoring provided high-resolution, real-time insight into reservoir behavior, allowing operators to observe fracture propagation, connectivity, and thermal response directly. Nanoparticle tracer technology provided independent verification of fracture connectivity and flow paths, significantly improving confidence in reservoir characterization. These diagnostic technologies will play a critical role in future geothermal developments by reducing uncertainty and enabling optimized stimulation and reservoir management strategies.



*Injector strain (top left - Stage4, top right - Stage 5) and temperature changes on fiber (bottom image) while pumping Stages 4 & 5 in 55A-29 and flow testing 55-29. Temperature is in degrees Fahrenheit and measured depth in feet*

The successful implementation of a propped EGS reservoir under superhot rock conditions represents a significant step toward commercial geothermal development. The ability to reliably create conductive fracture networks, confirm connectivity, and sustain circulation demonstrates the technical feasibility of scalable geothermal reservoir engineering in extreme environments. These results provide a clear pathway toward future multi-well geothermal developments capable of delivering high-temperature geothermal energy for baseload power generation.

## Field Testing and Demonstration of Closed-Loop Coaxial Geothermal Applications in Grimes County, Texas

*by Guizhong Chen, Chuck Wright, Alexis Garcia, Jessica Stump, NOV*

A modern drilling testing site was built in 2014 in Navasota, Grimes County, Texas. Since then, over 70 wellbores have been drilled and over 65 cement plug jobs have been conducted in the 200+ acre test facility. Most of the test wellbores were drilled deviated or horizontally for various fit-for-purpose test requirements. The test site is equipped with a 1500-HP drilling rig with the latest NOVOS drilling automation package, wired drill pipe (5-in. and 4-in.) and wired telemetry system, three drilling robots, and the driller's cabin situated on the ground level (Chen et al. 2025a). Open hole cement plugs, varying from 1 to 6 plugs at one job, have always been pumped to fulfill regulatory agreement, long-term wellbore and formation integrity, and repeated use of the same casing shoe for future open hole drilling tests (Chen et al. 2025b). Geomechanics studies have been performed to assure wellbore stability and safe drilling operations (Chen et al. 2023).

In addition to the three early-drilled surface hole locations (Well 58, Well 71, and Well 40 following the sequence of the spud time), a 11,500-ft vertical geothermal test well (Well 70) was drilled in 2020 (Hewlett et al. 2021), with a distributed temperature system (DTS) fiber-optic cable permanently cemented behind the 6-in. production casing from surface to the casing shoe. A surface DTS interrogator is installed to allow for continuously accessing the downhole temperature data at every foot of wellbore (Chen et al. 2024). The geothermal test well is completed with a full string of a vacuum-insulated-tubing (VIT, 4-in. x 2-7/8-in. with 2.441-in. inner diameter) and it appears to be the deepest VIT install in the world for a single-well, coaxial, closed-loop geothermal application based on a recent ChatGPT inquiry.

The geothermal test well indicates a 278°F (136.7°C) static formation temperature at 11,500 ft, confirmed by conventional and wired measurement-while drilling (MWD) tool measurements and fiber-optic DTS measurements during well shut-in (Chen et al. 2024; Caines, 2025). The calculated geothermal gradient of 1.8°F/100ft or 32.8°C/km (Chen et al. 2025c) is consistent with the areal analysis (NETL, 2019).

A series of flow testing has been conducted with the duration of 5.5 to 24 hours and 150, 75 and 37.5 gal/min flow rates using filtered fresh water. Surface temperature up to near the boiling point at low rate was observed and thermal power of ~1.5 mega-watts (MWth) was calculated at a high rate at the end of the testing period (Chen et al. 2026). Additional thermal modeling has been performed and calibrated against the field data (Chen et al. 2022; Aljubran et al. 2025; Zhang et al. 2026a). A techno-economic analysis thermal energy storage at the test site using a deep, open-loop doublet in the permeable Austin Chalk formation has been conducted (Wu et al. 2026).

An Organic-Rankine-Cycle (ORC) power generation demonstration feasibility has been assessed using the drilling test facility and the geothermal test well. The project is currently undergoing budgetary review. The study shows that the geothermal test well is capable of producing marginal electric power using a 75-MWe nameplate ORC unit. An incline circulating heater at surface is considered as a contingent add-on to assure the success of the demonstration (Thinkgeoenergy, 2025; Chen et al. 2026).

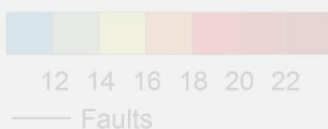
The uniqueness of the drilling test facility can also prove the capability of testing innovative critical mineral extraction, nuclear waste canister disposal, offshore riser handling technology, aquifer thermal energy storage (ATES), data center powering and cooling technology, automated managed pressure drilling (MPD) and automated managed temperature drilling (MTD) technologies (Zhang et al. 2026b). All the above-mentioned technologies have gone through cautious project framing and rigorous technical review.

## Field Testing and Demonstration of Closed-Loop Coaxial Geothermal Applications in Grimes County, Texas (Cont.)

*by Guizhong Chen, Chuck Wright, Alexis Garcia, Jessica Stump, NOV*

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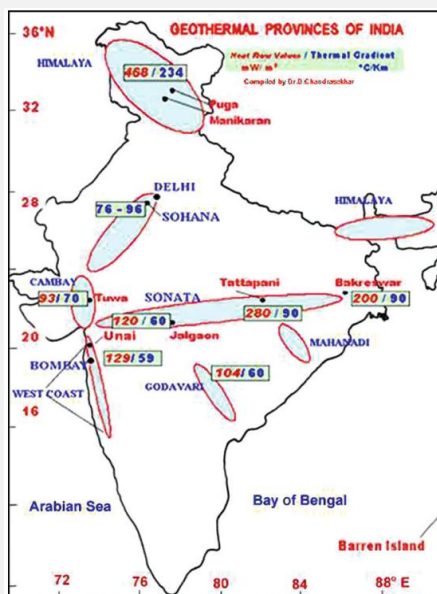
## STUDENT ARTICLE

### Unlocking the Earth's Heat: A Geoscientific Perspective on the Geothermal Potential of the India's Pranhita-Godavari Basin

**Siddhi Anil Gad Kerkar (MSc PTech) & Dr. Sagar Thakurdas (Asst. Professor)**  
**Nowrosjee Wadia College, Pune (Student Chapter No. 6206)**

India's energy demand to supply ratio has been consistently rising due to growing population needs, urbanization and massive industrialization. Considering this, geothermal energy is a new avenue in clean energy space that has the potential to provide a constant, reliable supply of baseload power derived from the Earth's natural internal heat. While high-temperature resources are typically found near active tectonic boundaries, modern exploration is increasingly uncovering immense geothermal potential within India's unique geological framework.

Several studies in the past suggest substantial geothermal potential in India's category 2 & category 3 sedimentary basins, with the Geological Survey of India identifying over 340 hot spring locations across the country. These geothermal manifestations are grouped into several major provinces (Fig. 1). For instance, the Himalayan Province boasts high-enthalpy systems like the Puga Valley and Manikaran, where reservoir temperatures can exceed 200 °C, driven by post-Tertiary granite intrusions. In central India, the SONATA (Son-Narmada-Tapi) lineament features the highly promising Tattapani field, which exhibits exceptional heat flow and reservoir temperatures up to 217 °C. Additionally, the Cambay graben and West Coast provinces in western India present significant moderate-enthalpy resources overlying thick sedimentary and basaltic formations.



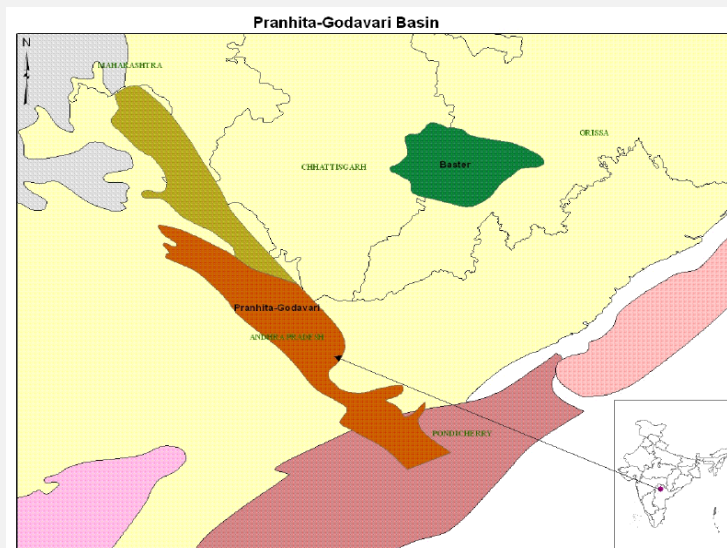
**Fig 1: Geothermal Provinces of India**

Among these provinces, **Pranhita-Godavari Basin** is recently gaining consensus because it serves as an **exceptional geoscientific model for a non-volcanic, passive intra-cratonic rift system**.

## STUDENT ARTICLE

### Unlocking the Earth's Heat: A Geoscientific Perspective on the Geothermal Potential of the India's Pranhita-Godavari Basin (Cont.)

*Siddhi Anil Gad Kerkar (MSc PTech) & Dr. Sagar Thakurdas (Asst. Professor) Nowrosjee Wadia College, Pune (Student Chapter No. 6206)*



**Fig 2: Geothermal Provinces of India**

**The Pranhita-Godavari Basin (Fig. 2)** is a major northwest-southeast trending tectonic lineament and continental rift zone situated on the Proterozoic platform of the Indian shield. The basin is characterized by a thick sequence of Gondwana sediments comprising sandstones, shales, and coal seams that act as an insulating blanket over the crystalline basement. From a structural standpoint, the basin is delineated by master faults and divided into sub-basins (such as the Chintalpudi, Godavari, and Chandrapur) separated by transverse transfer zones like the Bhadrachalam and Wamanapalli ridges. **These deep-seated faults play a critical role in facilitating deep fluid circulation, which is the primary mechanism for transferring subsurface heat to the surface.**

Detailed subsurface geophysical and thermal studies indicate that the central Godavari basin is a **moderate-enthalpy geothermal system with exceptional heat flow characteristics**. Thermal logging of exploratory boreholes drilled up to 1,000 meters deep reveals highly anomalous thermal regimes. **The geothermal gradient in areas like the Manuguru field ranges from 22.5 to 105.5 °C/km, while the heat flow varies between 83 and 388 mW/m<sup>2</sup>.** These values are significantly higher than the global average for continental crusts, directly reflecting the radiogenic decay of uranium and thorium in the underlying granites.

As energy demands shift toward sustainable solutions, integrating multi-disciplinary studies is crucial for unlocking geothermal resources especially for a growing economy like India. This can help in accelerating exploitation schemes by accurately mapping the fluid pathways and heat sources, thus making these moderate-enthalpy basins highly scalable for a clean energy future.

## GEOTHERMAL NEWS

### Energy Department Announces \$171.5 Million To Expand U.S. Geothermal Energy

The U.S. Department of Energy (DOE) today announced a funding opportunity of \$171.5 million to support next-generation geothermal field-scale tests for both electricity generation and exploration drilling to support characterization and potential confirmation of promising geothermal prospects.

[Energy.gov](https://www.energy.gov)

February 25, 2026

*Support for Field-Scale Tests and Exploration Drilling Can Help Advance Affordable, Reliable, Secure Geothermal Energy for American Homes and Businesses*

**WASHINGTON**—The U.S. Department of Energy (DOE) today announced a funding opportunity of \$171.5 million to support next-generation geothermal field-scale tests for both electricity generation and exploration drilling to support characterization and potential confirmation of promising geothermal prospects. The activities enabled by this opportunity will help deliver on President Trump's Executive Order, [Unleashing American Energy](#) by advancing geothermal technology, innovation, and exploration, in turn supporting the potential for geothermal energy to provide affordable, reliable, around-the-clock domestic electricity to Americans nationwide.

"Work under this opportunity will directly support our commitments to advance energy addition, reduce energy costs for American families and businesses, and unleash American energy dominance and innovation," said **DOE Assistant Secretary of the Hydrocarbons and Geothermal Energy Office Kyle Haustveit**. "Thanks to President Trump's America First Energy Agenda, these demonstrations and drilling activities will help us realize the enormous potential of geothermal to spur domestic manufacturing, enable data center growth, and provide affordable, reliable, and secure energy solutions nationwide."

The funding opportunity includes six topics with varied levels of funding and awards anticipated. For the first round of applications, two of the six topics will be open, seeking field tests for enhanced geothermal systems and drilling for next-generation and hydrothermal resource characterization / confirmation.

Although the United States leads the world in geothermal electricity capacity with about four gigawatts, DOE analysis shows the potential for at least 300 gigawatts of reliable, flexible geothermal power on the U.S. grid by 2050. Projects under this opportunity are expected to help derisk geothermal development approaches and locations nationwide, which can encourage private investment, spur industry growth, and help realize the country's geothermal potential.

Letters of Intent for the opportunity are due March 27, 2026, and full applications are due April 30, 2026. Learn more about the Notice of Funding Opportunity topic areas, expected funding, and how to apply [here](#).

*Learn more about geothermal research in DOE's Office of Geothermal [here](#).*

## GEOHERMAL NEWS

### GEL achieves double first for UK with production of geothermal electricity and zero-carbon lithium on a commercial scale

Published 3rd March 2026

On 26 February GEL started delivering the UK's first ever geothermal electricity 24/7 from its United Downs plant in Cornwall, marking a huge breakthrough for the nation's renewable energy mix and for UK geothermal power. Octopus Energy has signed a long-term deal to purchase at least 3 MW of constant, weather-independent power from United Downs – enough to supply 10,000 homes.

GEL has two other deep geothermal sites currently under development in Cornwall which will deliver a further 10 MW of base-load geothermal power in the UK by 2030. The British Geological Survey has calculated that more than 200 GW of thermal heat, hot enough to generate power, could be produced from onshore geothermal sources. That is the equivalent of over 100 nuclear power stations.

In another UK first, GEL has started commercial scale production of zero-carbon lithium carbonate, the key raw material used in the production of rechargeable batteries, including those that power electric vehicles and energy storage systems. From its February 2026 starting point, GEL the plant has the capacity to produce 100 tonnes per annum.

The geothermal fluid at United Downs that is used for the power plant comes from the deepest well ever drilled on UK soil at over 5km deep well. The water brought to the surface contains over 340 parts per million (ppm) of battery-grade lithium carbonate equivalent (LCE) making it one of the highest concentrations of lithium in the world. GEL aims to scale its production to over 18,000 tpa over the next decade, enough for circa 250,000 EV batteries a year – 65% of the UK's 2024 Battery Electric Vehicle (BEV) registrations.

Read more details [here](#).



*The United Downs geothermal power plant in Cornwall, UK (source: GEL)*

## GEOHERMAL NEWS

### Project Innerspace and SPE launch the Geothermal Resources Management System

Project InnerSpace and SPE have launched the Geothermal Resources Management System (GRMS), a new standard based on SPE's Petroleum Resources Management System, designed to provide capital markets with a clear and consistent framework for classifying, reporting, and evaluating geothermal resources.

Read the full article: [Link](#)



## SPE GTTS ACTIVITIES: WEBINARS

### Geothermal Energy

Gabrijel Grubac, SPE GTTS Vice-Chair, was invited to speak by the SPE Lagos Section and the SPE FUTA Chapter, where he delivered valuable insights on Enhanced Geothermal Systems and highlighted the future opportunities for geothermal development.



SOCIETY OF PETROLEUM ENGINEERS  
FUTA CHAPTER

PRESENTS

**Technical Session**

TOPIC:

### GEOTHERMAL ENERGY:

From Crust to Power

**About the Speaker**

Gabrijel Grubac is currently a Senior Completions Engineer at Mazama Energy, where his work is focused on Enhanced Geothermal Systems (EGS) and SuperHot rock geothermal development. He is a completions and stimulation engineer with over a decade of experience in the global energy industry, working across the United States, Europe, and the Middle East in onshore and offshore assets. His current focus lies in enhancing the completion of the future for SuperHot rock and help build systems that aim to achieve uniformity in stimulation and heat harvesting to ultimately enhance output and economics of projects. Gabrijel is also actively engaged in geothermal and energy education through SPE, supporting technical knowledge transfer and workforce development for next-generation geothermal systems and beyond.



**Gabrijel Grubac**  
Senior Completion Engineer at Mazama

Speaker

Google Meet 26TH FEB 2026 6PM Thursday

in f @ @SPE FUTA



Solutions.  
People.  
Energy.™

**SPE Lagos YP  
Technical Series**

THEME:

### GEOTHERMAL ENERGY:

HARNESSING THE EARTH'S HEAT FOR A  
SUSTAINABLE FUTURE

**SPEAKER**



**GABRIJEL GRUBAC**  
Senior Completions Engineer  
Mazama Energy

**MODERATOR**



**FRANKLYN OKAFOR**  
Geoscientist  
Dagobere Nigeria Limited

Monday,  
Dec. 1st, 2025  
5 PM

ZOOM REGISTRATION LINK:  
[https://bit.ly/SPEL\\_DecYPTM](https://bit.ly/SPEL_DecYPTM)

**Obianuju Igboke**  
Section Chair

**Elesie Ezenwanne**  
Programs Chair

**Kolade Owolala**  
YP Chair

f X SPE Lagos SPE Lagos Section 61 SPE Lagos Section - Young Professionals (YP) speilagos.org

### Enhanced geothermal System >300°C: the Newberry Volcano path to SuperHot Rock geothermal

**Moderator: Alexander Richter,**  
*ThinkGeoEnergy*

**Speaker: Gabrijel Grubac,**  
*Mazama Energy*

World's first successful EGS exceeding 300 °C, with insights from the Mazama Energy Newberry Volcano project in Oregon, USA.



**Recording link:**

<https://www.youtube.com/watch?v=3L3694VMIWI>

### Focus on Geothermal

Energy for the Weekend



16 January 2026

14:00 CET

08:00 ET

**Enhanced Geothermal  
System >300°C:  
The Newberry Volcano Path  
to SuperHot Rock Geothermal**



ALEXANDER  
RICHTER  
THINKGEOENERGY



GABRIJEL  
GRUBAC  
MAZAMA  
ENERGY



## SPE GTTS ACTIVITIES: WEBINARS

### End-to-end Integrated Geothermal System modeling with Volsung



#### SPECIAL GEOTHERMAL TALKS

January 2026

#### END-TO-END INTEGRATED GEOTHERMAL SYSTEM MODELING WITH VOLSUNG

JANUARY 29<sup>TH</sup> | 12:00 PM – 1:00 PM | US CENTRAL TIME | [ZOOM WEBINAR](#)

**DR. ESTEBAN GÓMEZ-DÍAZ**

GEOTHERMAL SPECIALIST, SEEQUENT



**Recording link:**

→ [SPEcial GTTS Webinar January 2026: End-to-end Integrated Geothermal System Modeling with Volsung.](#)

In this webinar **Esteban Gómez-Díaz** explored Volsung, purpose-built geothermal simulation solution. Volsung unifies reservoir, wellbore, and surface-network modeling into a single, intuitive platform. By simulating fluid and heat flow, well performance, and operational impact in one environment, engineers can capture the critical feedback loops essential for accurate forecasting and sustainable field development.

### Risk and Uncertainty Assessment of Geothermal Projects using the reverse Enthalpy Methodology



#### SPECIAL GEOTHERMAL TALKS

February 2026

#### RISK AND UNCERTAINTY ASSESSMENT OF GEOTHERMAL PROJECTS USING THE REVERSE ENTHALPY METHODOLOGY

FEBRUARY 26<sup>TH</sup> | 12:00 PM – 1:00 PM | US CENTRAL TIME | [ZOOM WEBINAR](#)

**ROBERTO GAMBINI**

CO-CEO, GEOLOGICAL ENGINEERING NETWORK SRL



**Recording link:**

→ [SPEcial GTTS February 2026 webinar: Risk and Uncertainty Assessment of Geothermal Projects \[...\]](#)

**Roberto Gambini** did an outstanding presentation on how do we turn complex subsurface data into confident investment decisions. He explained the Reverse Enthalpy Methodology (REM), an innovative approach tailored to the unique realities of geothermal development of showing the audience the potential of superhot rock but also having a sober approach to the challenges that lie ahead.

## SPE GTTS ACTIVITIES: WEBINARS

Why the oil and Gas Sector is a key player in Building on 50 years of EGS Progress?



**SPECIAL GEOTHERMAL TALKS**  
March 2026

**WHY THE OIL AND GAS SECTOR IS A KEY PLAYER IN BUILDING ON 50 YEARS OF ENHANCED GEOTHERMAL SYSTEMS PROGRESS**

MARCH 26<sup>TH</sup> | 12:00 PM – 1:00 PM | US CENTRAL TIME | [ZOOM WEBINAR](#)

**ANGELA SELIGMAN**  
SENIOR GEOSCIENTIST - CLEAN AIR TASK FORCE (CATF)



 **Recording link:**  
[→ SPEcial GTTS March 2026 webinar: Why the oil and gas sector is a key player in building on 50 years of EGS progress](#)

**Angela Seligman** in her presentation answer the main question about: What is the difference between conventional and next-generation geothermal and why does it matter?; What overlaps exist between expertise in the geothermal and O&G sectors?; What progress has been made over the last 50 years in enhanced geothermal systems development?; What additional needs are needed to unlock geothermal's full potential?

### SPE Live Why Geothermal? Why Now? What is Driving the New Geothermal Era?

Why is geothermal suddenly everywhere in the energy conversation? In this SPE Live discussion, the speakers exchanged perspectives on what makes this moment particularly interesting: geothermal builds on decades of expertise in drilling, reservoir engineering, and field operations. Many of these capabilities already exist.

Monday, 30 March | 0900 CDT (UTC-5)

**SPE Live:**  
**Why Geothermal? Why Now?**  
**What is Driving the New Geothermal Era?**

**LIVE**







KATERINA YARED    SORAYA BORNAIZ    CAITLY SMITH    JOHANNA OSTRUM

**Follow us online for more webinars to be announced during 2026**

## SPE GTTS ACTIVITIES: CONFERENCES

### Stanford Geothermal Workshop California, USA. February 9-11, 2026

The SPE Geothermal Technical Section (GTTS), represented by [add names], proudly took part in the recent Stanford Geothermal Workshop. Our participation reinforced the value of staying actively engaged with the broader geothermal community, where the exchange of ideas and continuous collaboration are essential to advancing the sector. The workshop offered an excellent platform to reconnect with long-standing colleagues, establish new connections, and gain fresh perspectives through insightful presentations and open technical discussions. Strengthening these relationships is key to driving innovation and ensuring we remain at the forefront of geothermal development.

Link to conference:

<https://lnkd.in/dCNiSbjn>



## SPE GTTS ACTIVITIES: CONFERENCES

### Stanford Geothermal Workshop California, USA. February 9-11, 2026

**Gabrijel Grubac** contributed to the workshop through technical work focused on Enhanced Geothermal Systems (EGS), including insights from a first-of-its-kind high-temperature EGS demonstration project. The project, presented at the workshop, highlighted the challenges and innovative solutions associated with creating and monitoring stimulated reservoir volumes in extreme geothermal environments.



His participation reinforces the importance of continuous collaboration and knowledge sharing within the geothermal community, especially as the sector advances toward more complex, high-enthalpy developments. We appreciate the dedication and technical leadership demonstrated by Gabrijel Grubac, whose work contributes to strengthening industry connections and promoting innovation across the field.

#### Implementation of the World's First Greater Than 300 C Propped EGS Reservoir

Link to paper: [Implementation of the World's First Greater Than 300 C Propped EGS Reservoir](#)

Wadood El-Rabaa, Alain Bonneville, Idris Ben-Fayed,  
Alexa Gonzalez, Geoffrey Gullickson, Oswaldo Perez



# SPE GTTS ACTIVITIES: CONFERENCES

Stanford Geothermal Workshop  
California, USA. February 9-11, 2026



**Pejman Shoeibi** had the opportunity to present ongoing work on the digitalization and use of AI for geothermal system operations, a topic that continues to gain relevance across the sector.

Pejman's contribution not only showcased innovative approaches but also helped strengthen connections within the geothermal community, reinforcing the importance of collaboration and knowledge sharing as key drivers of progress in the field.



## SPE GTTS ACTIVITIES: CONFERENCES



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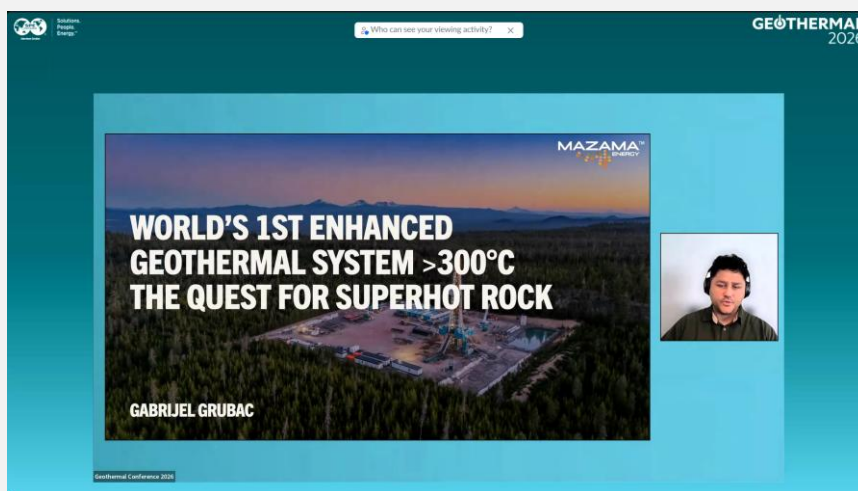
Accelerating Geothermal Energy

11–12 March 2026 | Hybrid Event | The Chester Hotel, Aberdeen

GEOTHERMAL  
2026

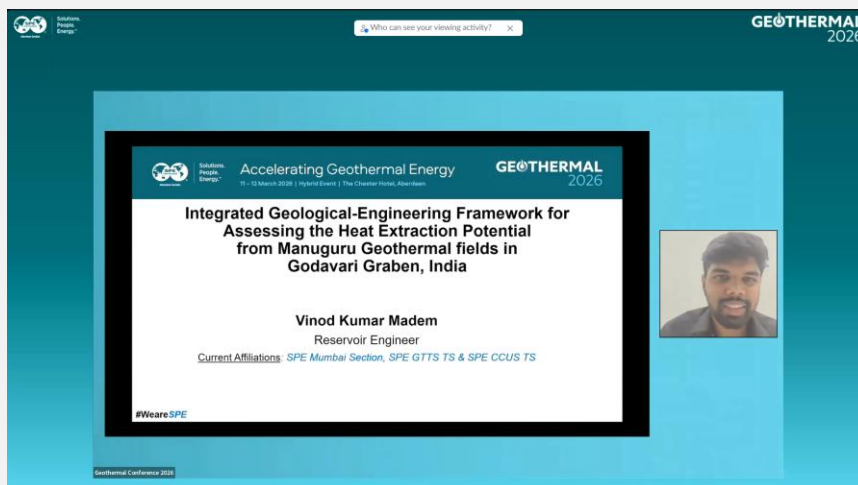
### WORLD'S 1ST ENHANCED GEOTHERMAL SYSTEM > 300 °C THE QUEST FOR SUPERHOT ROCK

Presenter: **Gabrijel Grubac**, Vice Chair, SPE GTTS



### INTEGRATED GEOLOGICAL-ENGINEERING FRAMEWORK FOR ASSESSING THE HEAT EXTRACTION POTENTIAL FROM MANUGURU GEOTHERMAL FIELDS IN GODAVARI GRABEN, INDIA

Presenter: **Vinod Kumar Madem**, Membership Chair (India), SPE GTTS



## SPE Upcoming Conferences

### SPE Europe Energy Conference & Exhibition.

ISTANBUL, TÜRKİYE. June 23-25, 2026

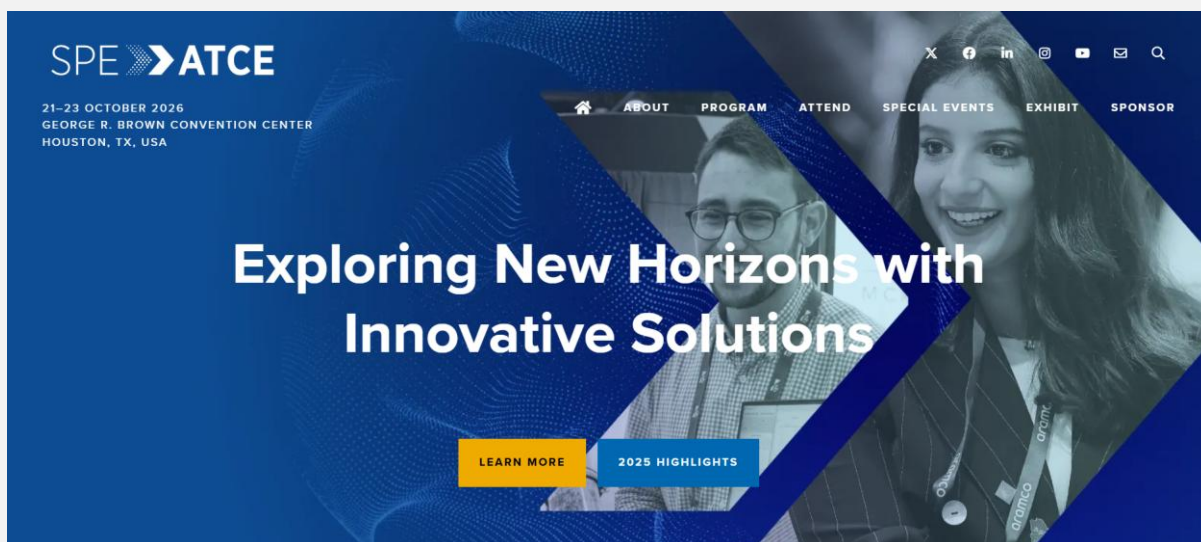
Link to conference: <https://www.spe-events.org/europeenergyconference>



### SPE Annual Technical Conference and Exhibition (ATCE)

Houston, Texas, USA. October 21-23, 2026

Link to conference: <https://www.atce.org/>



## GEOHERMAL CONFERENCES

Click on the title of each conference to get more information

### April, 2026

**Geothermal Investment Forum** - April 7 - 8, 2026 - Houston, Texas

**European Geothermal PhD Days 2026** - April 8 - 10, 2026 - Leoben, Austria

**Global Sustainable Islands Summit 2026** - April 20 - 22, 2026 - Gran Canaria, Spain

**XXXII Annual Geothermal Congress / Mexican Geothermal Congress** - April 23 - 24, 2026 - Puebla, Mexico

**Mine Water Geothermal Energy Symposium** - April 29 - 30, 2026 - Newcastle, England (virtual participation possible)

### June, 2026

**International Underground Thermal Energy Storage Symposium** - June 7, 2026 - Calgary, Canada

**World Geothermal Congress 2026** - June 8 - 11, 2026 - Calgary, Canada

**2026 EAGE Annual Conference & Exhibition** - June 8 - 11, 2026 - Aberdeen, UK

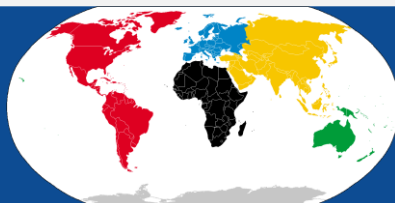
**IGC Invest Geothermal Conference** - June 17 - 18, 2026 - Frankfurt, Germany

**US Rock Mechanics / Geomechanics Symposium** - 21-24 June 2026 - Tucson, Arizona

### July, 2026

**15th ITB International Geothermal Workshop** - July 13 - 14, 2026 - Bandung, Indonesia

**SPE Africa Geothermal Workshop** - July 20 - 21, 2026 - Nairobi, Kenya



## GEOTHERMAL CONFERENCES (Cont.)

Click on the title of each conference to get more information

### August, 2026

**NZ Geothermal Week** - August 10 - 15, 2026 - Taupo, New Zealand

**12th Indonesia International Geothermal Convention & Exhibition** - August 19 - 21, 2026 - Jakarta, Indonesia

### September, 2026

**Budapest Geothermal Energy Summit** - September 14, 2026 - Budapest, Hungary

**Geothermal Rising Conference** - September 20 - 23, 2026 - Houston, Texas

**European Geothermal Energy: exploration, Drilling & Reservoir Management** - September 8 - 9, 2026 - Berlin, Germany

### October, 2026

**Praxisforum Tiefen Geothermie** - October 14 - 16, 2026 - Pullach, Germany

**11th African Rift Geothermal Conference (ARGeo C11)** - October 19 - 24, 2026 - Moroni, Comoros

**German Geothermal Congress / Der Geothermie Kongress** - October 20 - 22, 2026 - Potsdam, Germany

### November, 2026

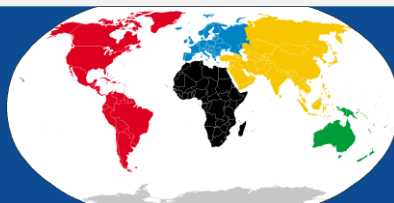
**International Geothermal Investment Summit** - November 2026 - Cappadocia, Türkiye

**EAGE Global Energy Transition** - November 2 - 6, 2026 - Hannover, Germany

**IADC Geothermal Drilling Conference & Exhibition** - November 17- 18, 2026 - Copenhagen, Denmark

**CONNECT 2026** - November 19, 2026 - Bern, Switzerland

**International Geomechanics Conference** - 17 – 19 November 2026 - Chengdu, China



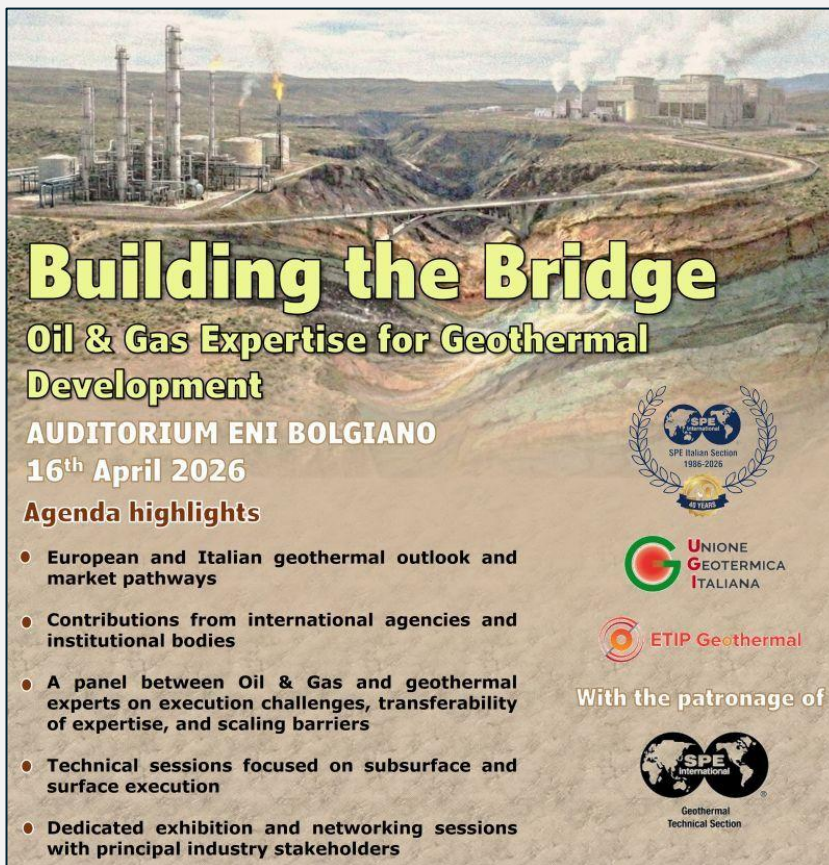
# COLLABORATION

## Building the Bridge: Oil & Gas Expertise for Geothermal Energy April 16th, 2026

SPE Italian Section along with SPE Geothermal Technical Section - GTTS, Unione Geotermica Italiana (UGI) and ETIP-Geothermal for coming together in pushing geothermal forward.

The event follows the publication of the SPE Italian Section technical bulletin “The New Wave of Geothermal Energy” and aims to continue that momentum by fostering discussion, networking and knowledge-sharing across the entire energy ecosystem.

A full day of keynotes, sharp dialogues, thematic sessions and an exhibition area designed to connect Oil & Gas capabilities with geothermal scaling.



**Building the Bridge**  
**Oil & Gas Expertise for Geothermal Development**

**AUDITORIUM ENI BOLGIANO**  
**16<sup>th</sup> April 2026**

**Agenda highlights**

- European and Italian geothermal outlook and market pathways
- Contributions from international agencies and institutional bodies
- A panel between Oil & Gas and geothermal experts on execution challenges, transferability of expertise, and scaling barriers
- Technical sessions focused on subsurface and surface execution
- Dedicated exhibition and networking sessions with principal industry stakeholders

 **SPE Italian Section 1995-2025 40 YEARS**

 **UNIONE GEOTERMICA ITALIANA**

 **ETIP Geothermal**

With the patronage of

 **SPE International Geothermal Technical Section**

**SPE GTTS is a proud supporter of this upcoming event and looks forward to this being another step to many more endeavors of amplifying geothermal energy through collaboration**

# COLLABORATION

## SPE Africa Geothermal Workshop July 20<sup>th</sup>-21<sup>st</sup>, 2026

### Powering Africa's Energy Future with Geothermal Innovation

The SPE Geothermal Technical Section is proud to jointly host the SPE Africa Geothermal Workshop 2026 with SPE Africa Region, taking place in Nairobi, Kenya, the heart of Africa's geothermal excellence.

Under the theme "Geothermal for Africa's Energy Future," this two-day workshop will bring together policymakers, financiers, operators, researchers, and innovators to explore how geothermal can become a pillar of Africa's clean energy transition



The poster features a blue and yellow color scheme. At the top, it displays the SPE Africa Region and SPE Geothermal Technical Section logos alongside the tagline "Solutions. People. Energy.™". The main title "SPE AFRICA GEOTHERMAL WORKSHOP 2026" is prominently displayed in large, bold letters, with "2026" in a yellow box. Below the title, the theme "THEME: Harnessing Africa's Geothermal Potential for a Sustainable Energy Future" is presented. The background of the poster shows a map of Africa and a geothermal power plant. At the bottom, the dates "20<sup>th</sup> - 21<sup>st</sup> JULY 2026" and the location "Fairmont The Norfolk Hotel NAIROBI, KENYA" are listed. A "REGISTER HERE" button with the URL <https://speafricageothermalworkshop.eventbrite.com> is provided. The hashtag #SPEAFRICAGEOTHERMALWORKSHOP and social media icons for LinkedIn, Facebook, Instagram, and X are also included.

**SPE AFRICA  
GEOTHERMAL  
WORKSHOP 2026**

**THEME**  
Harnessing Africa's Geothermal  
Potential for a Sustainable Energy Future

20<sup>th</sup> - 21<sup>st</sup> JULY 2026 | Fairmont The Norfolk Hotel  
**NAIROBI, KENYA**

**REGISTER HERE**  
<https://speafricageothermalworkshop.eventbrite.com>

#SPEAFRICAGEOTHERMALWORKSHOP

in f @ x SPE Africa Region

# COLLABORATION

## Expanding GTTS impact through cross-sectional engagement

The **GTTS Technical Directors team** is preparing to expand its engagement across SPE by strengthening collaboration with other Technical Sections. In the coming months, we aim to **leverage cross-disciplinary synergies, co-develop impactful events, and further position geothermal within the broader energy ecosystem.**

We invite all SPE Technical Sections interested in joint initiatives—whether workshops, webinars, or technical exchanges—to connect with us as we roll out this effort. GTTS is committed to building strong, collaborative partnerships that drive innovation and knowledge sharing across the industry.

If your section is interested in engaging, we welcome the conversation and look forward to working together.



### Technical Sections

|                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Artificial Lift</li> <li>Asset Management</li> <li>Carbon Dioxide Capture, Utilization and Storage</li> <li>Critical Minerals</li> <li>Data Science and Engineering</li> <li>Drilling Systems Advancement</li> <li>Drilling Uncertainty Prediction</li> <li>Flow Assurance</li> <li>Flow Measurement</li> <li>Geomechanics</li> <li>Geothermal</li> <li>Human Factors</li> <li>Hydraulic Fracturing</li> </ul> | <ul style="list-style-type: none"> <li>Hydrogen</li> <li>Integrated Reservoir Management</li> <li>Methane</li> <li>Offshore Completions</li> <li>Plug and Abandonment</li> <li>Research and Development</li> <li>Robotics and Autonomous Systems</li> <li>Separations Technology</li> <li>Sustainable Development</li> <li>Water Life Cycle and Strategy</li> <li>Wellbore Positioning</li> <li>Well Integrity</li> </ul> |
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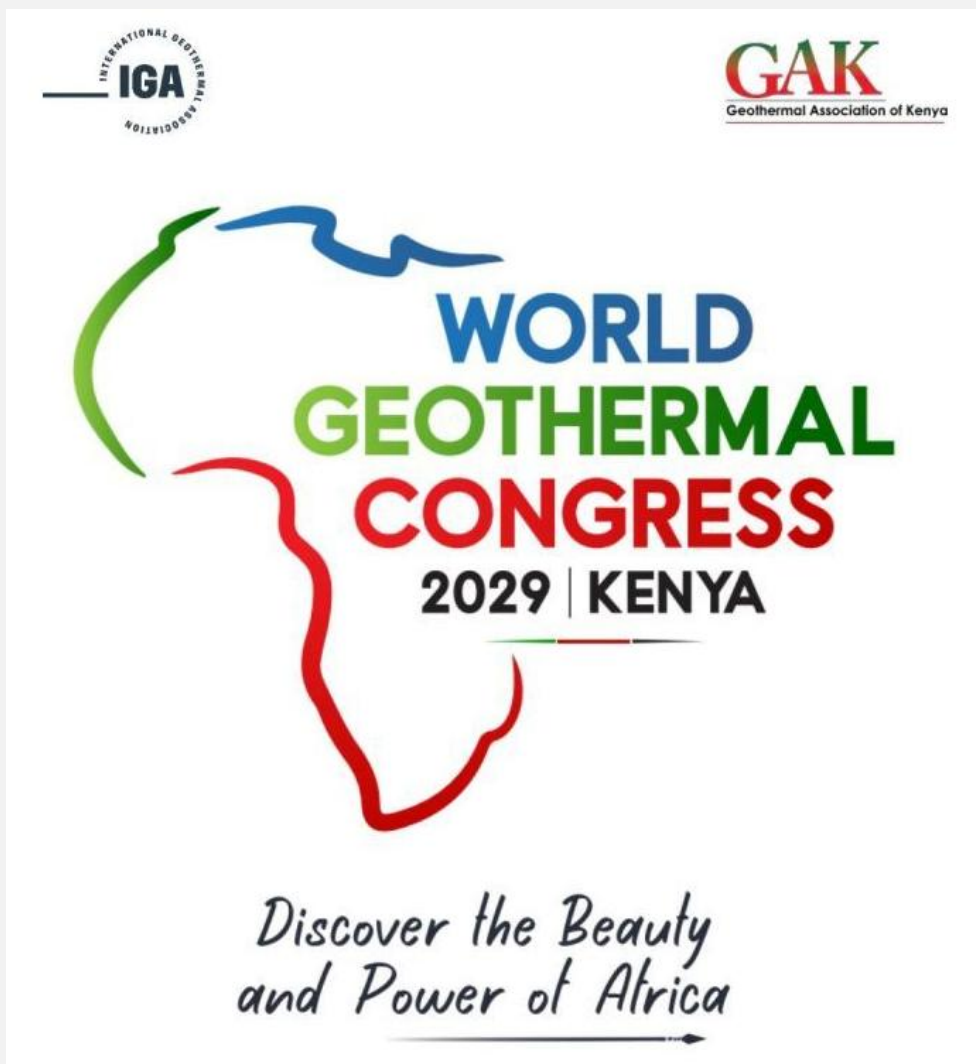
## GEOHERMAL CONFERENCES

Great news for geothermal energy in Africa!. **Kenya selected to host the World Geothermal Congress 2029.**

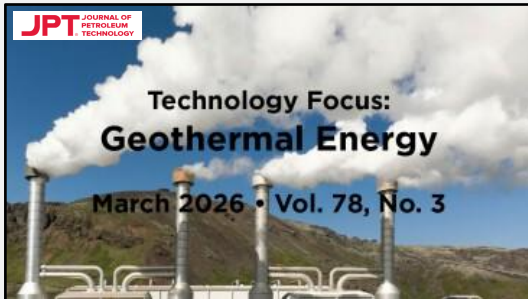
Nairobi, Kenya has been selected to host the World Geothermal Congress 2029, the first time this global event will be held on the continent. A huge milestone for the region and a strong recognition of Africa's leadership in geothermal development.

Congratulations to International Geothermal Association and Geothermal Association of Kenya (GAK) for spearheading it and enabling an increase in geothermal momentum.

Press release: <https://lnkd.in/d82p38Ps>



## FEATURED JPT GEOTHERMAL ARTICLES



**JPT** JOURNAL OF PETROLEUM TECHNOLOGY

**Technology Focus:  
Geothermal Energy**

March 2026 • Vol. 78, No. 3

**Geothermal Energy**

Geothermal development is gaining steam and entering a transformative era, driven by breakthroughs in adapting and improving on engineering, drilling, completion, and production technologies to the efficient extraction of heat from the Earth.

March 1, 2026 • Shaun Toralde • Journal of Petroleum Technology



**JPT** JOURNAL OF PETROLEUM TECHNOLOGY

**US Energy Department Announces \$171.5M for Geothermal Energy**

The US Department of Energy said it is making available \$171.5 million to support field-scale tests for geothermal electricity generation and exploration drilling.

March 5, 2026 • JPT Staff • Journal of Petroleum Technology

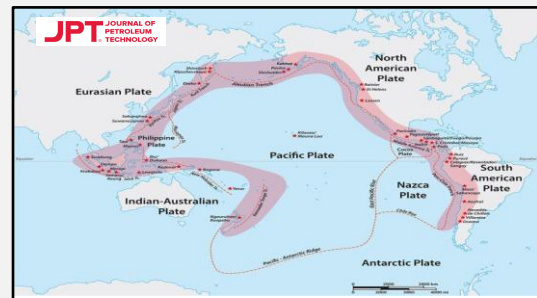


**JPT** JOURNAL OF PETROLEUM TECHNOLOGY

**US Universities Turn to Geothermal To Power the Next Generation of Buildings**

From Washington to Rhode Island, college campuses are tapping into geothermal energy to heat and cool new construction projects.

December 11, 2025 • Megan Couch • Journal of Petroleum Technology



**JPT** JOURNAL OF PETROLEUM TECHNOLOGY

**Japan Invests To Fire Up Geothermal Power**

As cutting-edge technologies unlock geothermal potential, Japan is channeling investments into powerhouse projects across the US, Indonesia, and New Zealand and priming its own reserves for a clean energy boom at home.

January 1, 2026 • Pat Davis Szymczak • Journal of Petroleum Technology



**JPT** JOURNAL OF PETROLEUM TECHNOLOGY

**National Laboratory of the Rockies Releases Geothermal Market Report**

The NLR has released its 2025 US Geothermal Market Report, documenting 4 years of industry growth and providing policymakers and stakeholders with an updated overview of the US geothermal market.

January 21, 2026 • JPT Staff • Journal of Petroleum Technology



**JPT** JOURNAL OF PETROLEUM TECHNOLOGY

**Guest Editorial: Optimizing Geothermal Performance With Artificial Lift Expertise**

Experience in subsurface production and lift design is shaping a new generation of geothermal operations built for reliability and scalability.

January 1, 2026 • Oghenekevwe S. Ovbije • Journal of Petroleum Technology

## FEATURED JPT GEOTHERMAL ARTICLES




**Advancements Bring Geothermal Power to the Surface**

From its origins running just a few light bulbs in Tuscany in 1904 to supporting baseloads on national power grids today, geothermal power generation has been driven by technological advancements. Many of these advancements stem from oil and gas exploration and production efforts.

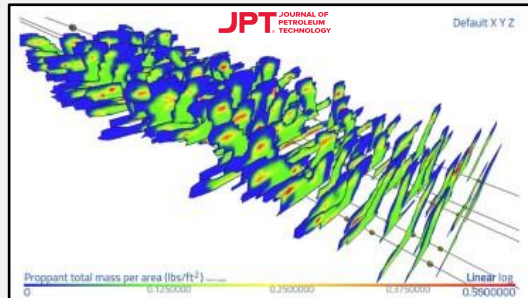
March 1, 2026 • Adam Wilson • Journal of Petroleum Technology




**Closed-Loop Geothermal Drilling Proves Effective in Romanian Pannonian Basin Basement**

This study evaluates the feasibility of drilling a closed-loop, unconventional geothermal system in the Pannonian Basin basement of Romania.

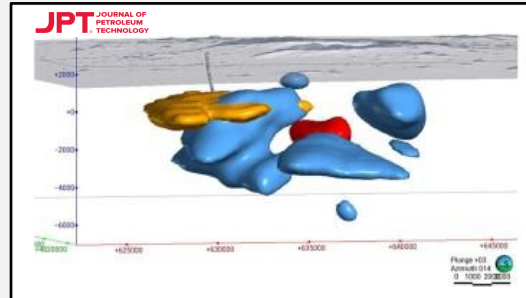
March 1, 2026 • Journal of Petroleum Technology

**Record-Breaking Enhanced Geothermal System Designed for Project Cape, Utah**

This paper describes modeling work performed to design fracturing treatment and spacing for wells in the Project Cape enhanced geothermal system in Utah.

March 1, 2026 • Journal of Petroleum Technology

**Enhanced Geothermal System Proppant Stimulation Targets High-Temperature Dry Rock**

This paper provides an account of the design, implementation, and operational insights from an enhanced geothermal system proppant stimulation targeting a volcanic, dry rock setting with an approximately 330°C bottomhole temperature.

March 1, 2026 • Journal of Petroleum Technology




**What Huff-n'-Puff Teaches Us About Geothermal Heat Recovery**

Switching from continuous circulation to cyclic huff-n'-puff operation in enhanced geothermal systems can significantly delay thermal breakthrough, sustain higher production temperatures, and improve long-term economic performance.

December 22, 2025 • Lokesh Kumar Sekar • Journal of Petroleum Technology




**Fervo Drills Its Hottest Well**

The appraisal well in Utah confirmed a geothermal resource exceeding 555°F at a depth of approximately 11,200 ft.

February 10, 2026 • JPT Staff • Journal of Petroleum Technology

# SPE Geothermal Technical Section Launches New Digital Platform With Mobile App for iOS and Android

Your SPE GTTS community is now available, anytime, anywhere. Explore events, news, and technical content on the new platform and app



What's on the updated GTTS webpage and app:

- Upcoming events and webinars: register quickly with just a few clicks
- Technical videos and recorded sessions: learn at your own pace
- Newsletters and announcements: stay informed on the latest updates
- Articles on geothermal, energy, and innovation: insights and analysis
- Board and leadership updates: news from GTTS leadership
- A simple feedback and contact form: share your thoughts or questions

To explore all these features and stay connected, check out the links below:

- [GTTS Website](#)
- [iOS app](#)
- [Android app](#)
- [GTTS SPE Connect Community](#)

## GET THE SPE GEOTHERMAL APP

Access all features on the go. Stay updated with the latest news, events, and technical resources from anywhere.



## EDITORIAL TEAM MESSAGE

### Witnessing Collaborative Efforts

#### *Fourth Edition of the SPE GTTS Newsletter*

**Geothermal Energy** is experiencing a remarkable global renaissance, stepping firmly out of its traditional geographic niches and into the broader spotlight of the energy transition. For decades, it has served as a reliable, baseload power source in specific volcanic regions.

Across the globe, we are witnessing unprecedented momentum. From breakthroughs in Enhanced Geothermal Systems (EGS) that are proving the viability of engineered reservoirs, to the deployment of innovative closed-loop systems and deep-drilling technologies, the boundaries of geothermal viability are expanding rapidly.

What is most exciting for our community is the profound crossover between traditional petroleum engineering and these next-generation geothermal innovations.

In this issue, we highlight the dynamic nature of our field, featuring recent insights shared by our board members and other subject matter experts to the geothermal community.

We invite you to dive into this issue, share your perspectives, and continue pushing the boundaries of what is possible in geothermal energy.

Thank you for your ongoing support and engagement.

### SPE GTTS Newsletter Editorial Team



**Sonia López Kovács**

SPE GTTS Membership  
Chair - Europe



**Vinod Kumar Madem**

SPE GTTS Membership  
Chair - India



**Mahmoud M. AlGaier**

SPE GTTS Membership  
Chair - Middle East

## GEOHERMAL RESOURCES



**Stanford | Doerr**  
School of Sustainability



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*Tell us what you think!*



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Energy.<sup>SM</sup>**



**Geothermal  
Technical Section**